Plant and Soil **191:** 279–290, 1997.

Soil and solution chemistry under pasture and radiata pine in New Zealand

R.L. Parfitt[1], H.J. Percival[1], R.A. Dahlgren[2] and L.F. Hill[1]
[1]*Landcare Research, PB 11052, Palmerston North, New Zealand* * *and* [2]*Department of Land, Air and Water Resources, University of California, Davis, CA 95616, USA*

Received 22 November 1996. Accepted in revised form 19 March 1997

Key words: acidification, carbon, carbon dioxide, nitrogen, pasture, *Pinus radiata*, sea salt

Abstract

The conversion of hill country pasture to exotic forest plantations is occurring rapidly (70,000 ha yr^{-1}) in New Zealand. Impacts of this land-use change on soil properties, soil fertility, and water quality are only beginning to be investigated. This study examines the effects of radiata pine (*Pinus radiata*) on soil and soil solution chemistry, in a region of low atmospheric pollution, 20 years after plantation establishment, assuming that the pasture and pine research sites had comparable soil properties before planting pine. The primary effects of conversion on soil chemistry were a decrease of organic carbon in the mineral soil that was balanced by an accumulation of the surface litter layer, a decrease in soil N, soil acidification, and increased pools of exchangeable Mg, K, and Na. Soil solution studies revealed a large input of sea salts by enhanced canopy capture of sea salts that contributed to much larger solute concentrations and elemental fluxes in the pine soil. Sea salts appear to accumulate in the micropores of pine soil during the dry summer period and are slowly released to macropore flow during the rainy season. This results in a progressive decrease in solute concentrations over the period of active leaching. While chloride originating from sea salt deposition was the dominant anion in the pine soil, bicarbonate originating from root and microbial respiration was the dominant anion in the pasture soil. Carbon dioxide concentrations in the soil atmosphere were 12.5-fold greater in the pasture soil than in the pine soil due to greater rates of root and microbial respiration and to slower diffusion rates resulting from wetter soil conditions in the pasture. Although elemental fluxes from the upper 20 cm of the soil profile were substantially greater in the pine soil, these losses were compensated for by increased elemental inputs resulting from nutrient cycling and enhanced canopy capture of sea salts.

Introduction

The conversion of hill country pasture to exotic forest plantations, predominately radiata pine (*Pinus radiata* D.Don), is occurring at a rate of approximately 70,000 ha yr^{-1} in New Zealand (Maclaren, 1996; Mead, 1995). Establishment of pine plantations is motivated by the favourable economic return for wood fibre, and by hillslope stabilization to protect the landscape. The conversion of pasture to forest is also being advocated as a method for attenuating New Zealand's net carbon dioxide emissions. While the conversion of pasture to pine may have several beneficial effects, the impact of this land-use change on soil properties, soil solution and water quality is only beginning to be investigated.

* FAX No: +6463559230. E-mail: parfitt r@landcare.cri.nz

Both solid-phase soil analyses and soil solution studies have been used to examine the effects of land-use change and ecosystem perturbations on soil properties and nutrient status. Investigations utilizing solid-phase samples commonly examine the surface 5 to 10 cm layer the soil layer most strongly affected by changes in vegetation (Dahlgren et al., 1991).

Previous studies of pasture conversion to pine have focused on changes to solid-phase soil properties. These studies indicate that pines generally increase levels of inorganic P, Olsen and Bray extractable P, mineralizeable N, nitrification potential, and sulfate-S compared to the previous pasture or grassland (Birk, 1992; Condron, 1996; Davis, 1994, 1995; Davis and Lang, 1991). Concomitantly, organic C, total N, organic P, and total P show a decrease following pine establishment. It is suggested that increased mineralization of organic matter and nutrient cycling (absorption of nutrients from lower horizons and return to the soil surface) are the primary mechanisms responsible for the observed changes in the nutrient status (Davis, 1995). Soil properties and nutrient pools may continue to change during the course of stand development; thus long-term impacts may not follow directly from the results of these short-term studies (Birk, 1992; Vitousek et al., 1989).

Since solid-phase nutrient pools are very large (100–1000's kg ha^{-1}, small changes in such large pools are not easily detected in the short-term, given the inherent spatial variability of soils on the landscape and the precision of analytical methods for solid-phase characterization. In contrast, soil solutions provide a powerful approach for detecting changes in soil processes due to perturbations of the ecosystem. Soil solutions play a vital role as a transporting agent and a solvent in soil formation, and provide information about current soil processes including acid/base chemistry, relative availability of nutrients, and leaching losses of nutrients. Previous studies have shown that differences in soil solution chemistry may be detected sensitively and rapidly after land-use changes or ecosystem perturbations (e.g. Cole and Gessel, 1965; Dahlgren and Driscoll, 1994; Dahlgren and Ugolini, 1989; Dahlgren et al., 1991; Rustad et al., 1993).

The primary objective of this study was to examine differences in soil properties between pasture and 20-year-old radiata pine growing on a former pasture using a combination of solid-phase and soil solution chemical analyses. These data are used to examine differences in acid/base chemistry, relative availability of nutrients, and leaching losses of nutrients between sites.

Materials and methods

Study site

The study site was located at the AgResearch Hill Country Research Station (Ballantrae) (170° 50' E; 40° 20' S), 35 km east of Palmerston North, New Zealand. The site is approximately 60 km from the ocean in a region of low athmospheric pollution which receives an input of sea salts, as observed by Heng et al. (1991) for another site in the area. Mean annual precipitation is 1291 mm and mean annual air temperature is 12.2 ° C (New Zealand Meteorological Service, 1983). The soils formed in greywacke sandstone/mudstone and are classified as the Wilford silt loam soil series; Typic Orthic Brown soils (Hewitt, 1992) or Typic Dystrochrepts (Soil Survey Staff, 1994). The soils investigated in this study occupy semi-stable convex ridges with slopes of 25 – 30°, northeast-facing aspects, and have an elevation of 165 m. Sites within a pasture and radiata pine (*Pinus radiata* D. Don) plantation were located on adjacent tracts approximately 300 m apart. Based on historical records and the morphological consistency of soil profiles in the pine and the pasture sites, we assume that the two adjacent sites had comparable soil properties before planting the pine trees. Thus any differences in current soil properties are believed to be primarily due to the change in land-use. The pasture consisted of low fertility grasses with some white clover, lotus, and suckling clover. The pasture site has the following fertilizer history:

	N	P	S
	(kg ha^{-1} yr^{-1})		
1972–80,	0	14	19
1981–87	0	11	15
1989	0	30	18
1992–96	14	19	16

The pasture site was grazed by sheep (1972–92) and bull beef (1992–96) at a stocking rate of 10.5 to 12 stock units ha^{-1} yr^{-1} The radiata pine were planted at 250 stems ha^{-1} in 2.9 ha of former pasture in 1975 and were 20 years old at the time of the study. The pine site was not grazed and received no fertilization.

Methods

Soil solutions were collected continuously for one water year when solutions could be sampled using tension lysimeters. Tension lysimeters (4 replicates) were installed beneath the major rooting zone at an average depth of 22 cm within both the pasture and pine sites. They consisted of a 25 mm diameter Millipore GFC filter with a 90 mm diameter glass fibre filter paper (GFC) wick (Stevenson, 1978; Stevenson and Beaumont, 1984), and lysimeters were installed in May 1995 by drilling holes into the hill slope at an angle of about 10° below horizontal. The 90 mm wick was close to a vertical position and extended over the 18–27 cm soil depth. The lysimeters were held at an average tension of 5 kPa; the tension after four weeks of continuous collection was approximately 2 kPa. Solutions were collected in glass flasks that were kept in the dark to prevent the growth of algae. The lysimeter sites within the pasture were protected by an electric fence that allowed stock to graze the pasture around the site but not tread on the soil surface immediately above the lysimeters.

Soil solutions were also collected in August by high-speed centrifugation using double-bottomed tubes at an average RCF of 2300 for 30 min. Soil samples were collected shortly after a rain event, when the soil was at field capacity, from the 0–10 and 10–20 cm depth of triplicate plots at each site. Within each plot, 15 soil cores were taken and composited into a single sample. Soils were transferred immediately to the laboratory, and soil solutions extracted within 2 h. About 25% of the soil solution was extracted by centrifuging the water content after centrifuging was very close to the water content measured at -100 kPa using a pressure plate. Therefore, the soil solution extracted by centrifugation is held in small pores (i.e. pore size diameters 0.003 – 0.06 mm) whereas the soil solution obtained by the tensiometer is held in macropores.

Rainfall volume was measured daily at a recording station 800 m from the study sites. Canopy throughfall volume in the pine forest was estimated from the daily rainfall less 26% interception measured for this site (F Kelliher, personal communication). Potential evapotranspiration was estimated from meteorological data and assumed to be similar for pasture and pine.

Precipitation, canopy throughfall from the pine, runoff from a lower concave slope in the pasture, and water from an intermittent stream draining the pine site were collected for selected precipitation events. They were sampled using plastic containers, which were filled with the samples, sealed and processed in the laboratory within one hour of collection.

Solution pH was measured immediately on non-filtered samples. Samples were filtered through a 0.2 μm membrane filter before analysis of cations and anions. Nitrate and ammonium were measured by autoanalyzer (Blakemore et al., 1987); chloride, sulphate, phosphate, and dissolved inorganic carbon by ion chromatography; and Ca, Ma, Na, and K by atomic absorption spectrometry. Bicarbonate concentrations were calculated from dissolved inorganic carbon concentrations and pH using equilibrium relationships (Dahlgren et al., 1997).

Solid-phase soil samples were collected in early October from six randomly selected plots in the vicinity of the lysimeter installations in both the pasture and pine sites. Bulk soil samples from the mineral soil were collected for the 0–10 and 10–20 cm depths; litter samples (Oi/Oe) were also collected from the pine site. Bulk density was determined by the coring method. Soil pH was determined after a one hour equilibration period in distilled-deionized water and 0.01 M $CaCl_2$ using a 1:2.5, soil:solution suspension. The pH of the ground litter samples was measured on a 1:5, litter:solution suspension. Organic carbon and total N were determined by dry combustion using a Leco induction furnace (FP–2000). Exchangeable cations and CEC were measured by 1 M NH_4Ac at pH=7 (Blakemore et al., 1987).

Soil atmosphere CO_2 concentrations were measured at the 10 and 20 cm depths by collecting a 1 mL sample of the soil atmosphere in a gas-tight syringe from four locations within both pasture and pine sites. Samples were collected by inserting a needle 5 cm into the side of a newly opened soil pit. Carbon dioxide concentrations were measured by gas chromatography using a thermal conductivity detector. Soil temperature and moisture were measured for each CO_2 sampling site. All sampling was performed within a two hour period at midday to eliminate diurnal effects.

Repeated measures ANOVA analyses was used to examine differences in solute concentrations between pine and pasture. A t-test procedure was used to examine differences in solid-phase soil properties between pasture and pine treatments and solute concentrations between soil solutions collected by centrifugation and tension lysimeters. All statistical analyses were performed using SYSTAT for Windows.

Results

Solid-phase soil properties

The most obvious difference between the two sites was an Oi/Oe litter layer in the pine forest ranging between 1–3 cm in thickness and having carbon and N masses of 6.3 and 0.1 kg ha^{-1} respectively (Table 1). This material had a C/N molar ratio of 53 and a relatively high pH (pH(H_2O)=5.35; pH($CaCl_2$)=5.02) for a litter layer suggesting a high base cation content.

Several differences in soil properties between the pine and pasture soils were found in the upper 20 cm of the mineral soil (Table 1). Mean soil pH was about 0.4 units lower throughout the upper 20 cm of the pine mineral soil for both water and $CaCl_2$ suspensions. While the bulk density was similar for the 0–10 cm increment, the bulk density of the 10–20 cm increment was 0.15

Table 1. Comparison of selected soil properties and nutrient pools (mean ± SEM) between soils under pasture and 20-year old raidata pine growing on a former pasture

	Pine			Pasture	
	Oi/Oe	0–10cm	10–20 cm	0–10 cm	10–20 cm
pH (H_2O)	5.35 (.06)	5.25 (.04)[b]	5.49 (.11)[d]	5.62 (.05)[b]	5.88 (.05)[d]
pH($CaCl_2$)	5.02 (.05)	4.44 (.06)[b]	4.63 (.10)[d]	4.82 (.04)[b]	4 .96 (.06)[d]
Organic C (g kg^{-1})	447.0 (17.8)	32.2 (3.9)[a]	15.9 (2.3)[d]	41.3 (1.9)[a]	28.3 (1.8)[d]
Total N (g kg^{-1})	9.8 (.3)	2.3 (.3)[b]	1.2 (.2)[d]	3.2 (.2)[b]	2.2 (.2)[d]
C/N molar ratio	53.3 (3.3)	16.5 (.6)[a]	15.2 (.6)	15.0 (.4)[a]	14.9 (.5)
CEC (cmol(+) kg^{-1})	ND[e]	17.1 (1.3)	12.9 (.9)	17.1 (.9)	14.8 (1.1)
Exchangeable cations					
Ca (cmol(+) kg^{-1})	ND	6.46 (.59)	5.15 (.47)[e]	7.41 (.75)	7.24 (.91)[c]
Mg (cmol(+) kg^{-1})	ND	2.20 (.10)[b]	1.79 (.11)	1.62 (.07)[b]	1.59 (.08)
K (cmol(+) kg^{-1})	ND	0.46 (.05)	0.24 (.02)	0.37 (.04)	0.31 (.05)
Na (cmol(+) kg^{-1})	ND	0.24 (.03)[b]	0.32 (.04)[d]	0.12 (.01)[b]	0.11 (.01)[d]
Base saturation (%)	ND	56.2 (4.7)	59.5 (5.2)	55.5 (2.4)	62.0 (3.0)
Bulk density (g cm^{-3})	ND	1.09 (.03)	1.27 (.07)[d]	1.05 (.06)	1.12 (.02)[d]
Nutrient pools					
Organic C (Mg ha^{-1})	6.3 (.9)	35.1 (4.2)[a]	20.1 (2.8)[d]	43.3 (1.8)[a]	31.7 (2.0)[d]
Total N (Mg ha^{-1})	0.1 (<.1)	2.5 (.3)[a]	1.6 (.2)[d]	3.4 (.2)[a]	2.5 (.2)[d]
Exch.-Ca (kg ha^{-1}	ND	1410 (127)	1315 (135)	1556 (148)	1622 (194)
Exch.-Mg (kg ha^{-1})	ND	291 (13)[b]	276 (19)[d]	207 (12)[b]	216 (11)[d]
Exch.-K (kg ha^{-1})	ND	197 (19)[a]	117 (13)	151 (15)[a]	136 (21)
Exch.-Na (kg ha^{-1})	ND	61 (7)[b]	93 (11)[d]	28 (3)[b]	28 (2)[d]

[a] and [b] : 0–10 cm layer is statistically different between pine and pasture at the p =0.10 and p=0.05 significance level, respectively.
[c] and [d] : 10–20 cm layer is statistically different between pine and pasture at the p=0.10 and p=0.05 significance level, respectively.
[e]ND = Not determined.

g cm^{-3} higher in the pine, which probably reflects the lower organic matter content. Organic carbon concentrations decreased in the pine treatment by an average of 9 and 12 g kg^{-1} in the 0–10 and 10–20 cm increments, respectively. Total N concentrations were 30 to 45% lower in the pine soil; however, N contents are confounded by the addition of 14 kg ha^{-1} yr^{-1} of N fertilizer to the pasture site during the 1992–96 period. The fertilizer may also have been partially responsible for the lower C/N molar ratio (p<0.10) in the 0–10 cm increment of the pasture soil (Table 1). Similar to C and N concentrations, pools of C and N were higher in the pasture soil. However, when the organic carbon pool was summed for the upper 20 cm of the mineral soil and the litter layer was included for the pine soil, the organic carbon pool was not statistically different (p >0.10) between the two treatments, indicating that the increase of organic carbon in the litter balanced decreases within the upper mineral soil. The cation exchange capacity and the base saturation were similar for both treatments; however, concentrations of exchangeable Na and pools of exchangeable Mg, K, and Na were generally greater in the pine soil. While the pool and concentration of exchangeable Ca were similar between sites, the addition of Ca through the historic application of superphosphate fertilizer, which contains about 20% Ca by weight, probably provides a residual effect on Ca concentrations in the pasture.

Ecosystem waterflows

The charge balance diagram representing mean concentrations over the water year indicates that total ion concentrations increase as the water flows through the hydrologic continuum: precipitation → canopy throughfall → soil solution → streamwater (Figure 1). Ionic concentrations in the precipitation were very low ($\leq$ 0.05 $mmol_c$ L^{-1}) for the study period and were dominated by Na and Cl with lesser concentrations of Ca, Mg, K, and SO_4. Throughfall from the pine canopy

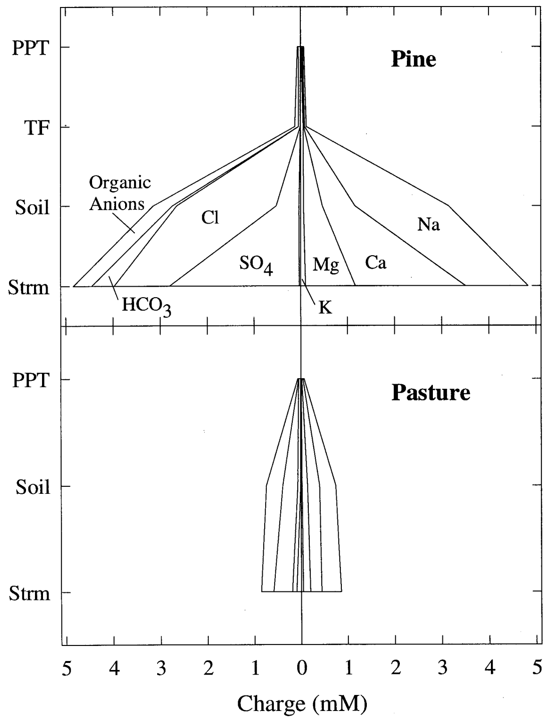

Figure 1. Charge balance diagram comparing the composition of precipitation (PPT), canopy throughfall (TF), lysimeter soil solutions (Soil), and streamwater (Strm) from pasture and pine sites. The anion charge deficit is calculated as the difference between the sum of measured inorganic cations and anions and is assumed to be due to dissociated organic acids. The order of ions in the pasture diagram is identical to that in the pine.

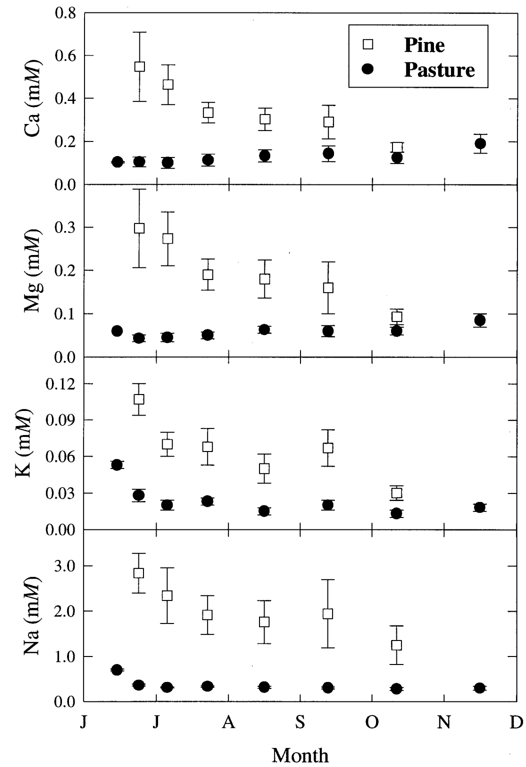

Figure 2. Concentrations (means ± SEM) of major cations for pasture and pine soil solutions collected by tension lysimeters.

showed an appreciable pH increase (5.8 to 6.3) along with a 6-fold enrichment of K and a 2-fold enrichment with Na; all other solute concentrations remained similar to those in the precipitation. Ionic concentrations in soil solution and stream runoff were about five times greater for the pine forest compared with the pasture. The major ions contributing charge in the soil solution and stream runoff were the Na, Ca, and Mg cations and the Cl, SO_4, HCO_3, and organic anions. The soil solution and streamwater composition were similar in the pasture, possibly suggesting that subsurface lateral flow emerging from the A horizons is the primary source of this streamwater. The pasture soil exceeded field capacity earlier and remained wetter later into the summer, resulting in nearly a two month water period of leaching and soil solution collection relative to the pine soil (Figures 2–5).

Concentrations of Ca, Mg, K, and Na were significantly ($p < 0.05$) greater in the pine soil solutions compared with the pasture (Figure 2). For a given sampling date, cation concentrations were 2– to 6–fold greater for the pine soil solutions. Among the cations, the abundance reported on a molar basis followed: Na >> Ca > Mg > K. Ammonium concentrations were less than detection limits (< 0.001 m*M*) for all soil solutions for the entire study period. All cation concentrations in the pine soil showed a decreasing trend over time during the study. In contrast, cation concentrations in the pasture soil were remarkably consistent throughout the study period, with the exception of the higher K and Na concentrations in the first collection.

Concentrations of Cl, SO_4, and NO_3 were significantly ($p < 0.05$) greater in the pine soil solutions, while HCO_3 concentrations were significantly ($p < 0.05$) greater in the pasture soil (Figure 3). Ortho-phosphate concentrations were less than detection limits (< 0.001 m*M*) in all soil solutions, in spite of the annual application of P fertilizer to the pasture soil. Organic anion concentrations, calculated as the difference between the sum of charge for measured inorganic cations and anions, were the same for both the pine (0.40 ± 21m*M*; mean ± SEM) and pasture (0.35 ± 0.02 m*M*; mean ±

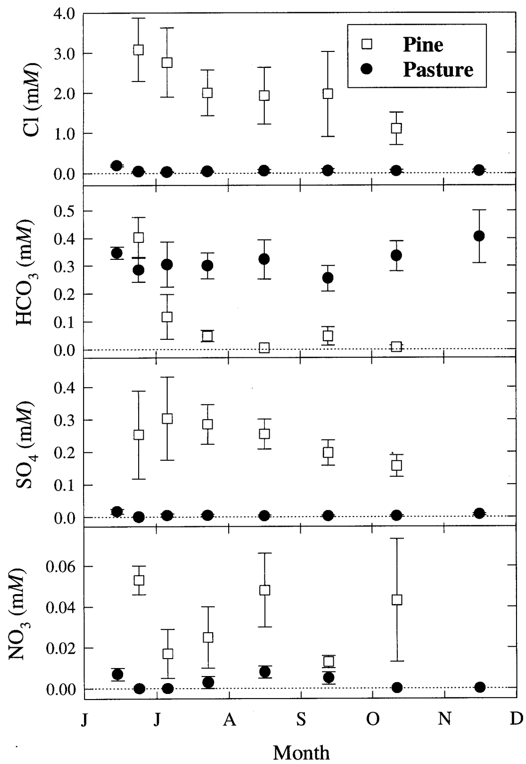

Figure 3. Concentrations (mean ± SEM) of major inorganic anions for pasture and pine soil solutions collected by tension lysimeters.

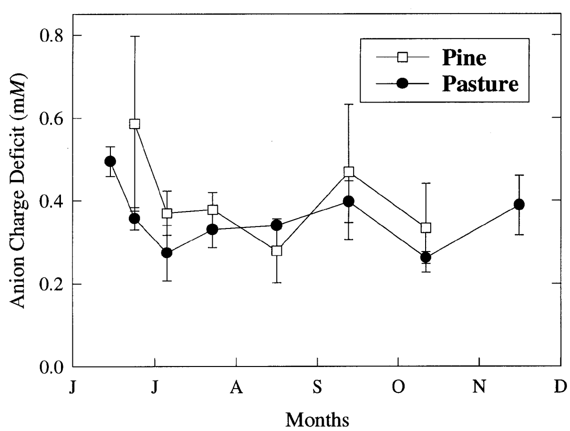

Figure 4. The anion charge deficit (mean ± SEM), assumed to be due to dissociated organic acids, for pasture and pine soil solutions collected by tension lysimeters.

SEM) soils (Figure 4). Among the anions, the abundance reported on a molar basis followed:

Pine Cl >> organic anions > SO_4> HCO_3 > NO_3,

Pasture HCO_3 > organic anions > Cl > SO_4 ∼ NO_3.

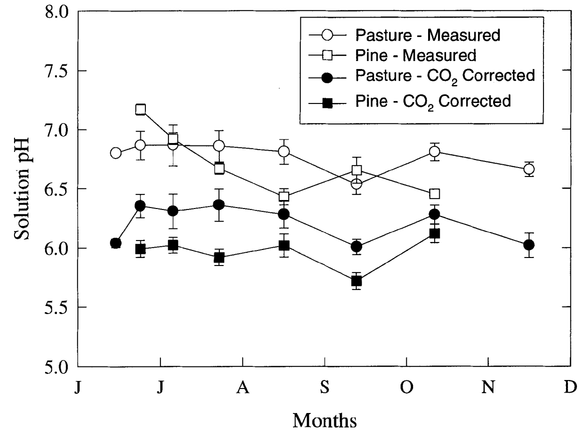

Figure 5. Solution pH (mean ± SEM) for pasture and pine site soil solutions collected by tension lysimeters. The soil solution pH corrected for CO_2 degassing in the tension lysimeters is also shown.

The most notable contrast is the dominance of Cl in the pine soil compared to HCO_3 in the pasture soil. Concentrations of Cl and SO_4 showed a decrease over time for the pine soil; however, HCO_3 and NO_3 concentrations in the pine soil along with all anion concentrations in the pasture soil showed no consistent trend over time.

The pH of soil solutions collected by tension lysimeters was similar for both the pine (6.70 ± 0.06; means ± SEM) and pasture soils (6.77 ± 0.04; means ± SEM) (Figure 5). Due to the large head space: soil solution volume ratio in the collection vessel of the tension lysimeter, appreciable CO_2 degassing of the soil solutions occurred before analysis. Carbon dioxide degassing results in a decrease in dissolved inorganic carbon concentrations. To maintain soil solution charge balance, the solution pH must increase to produce a greater dissociation of H_2CO_3 to HCO_3, thus maintaining a constant HCO_3 concentration and soil solution charge balance. As a result, the solution pH values do not accurately represent the in situ soil solution pH values. Based on measured values for the head space soil solution volume and soil solution pH and HCO_3 concentrations, the pH of the in situ soil solutions were calculated assuming equilibrium conditions following the methodology of Suarez (1987). The pH values obtained following correction for CO_2 degassing indicated a significantly ($p < 0.05$) lower pH value for the pine (5.97 ± 0.05; mean ± SEM) soil solutions compared with the pasture (6.2 ± 0.05; means ± SEM) (Figure 5). The pH values corrected for CO_2 degassing showed no discernible trend with time over the study period.

Elemental fluxes from the upper 20 cm of the soil profile were estimated from monthly measurements or estimates of precipitation, canopy interception, evapotranspiration, and solute concentrations of soil solutions collected by lysimetry. Fluxes were appreciably higher in the pine soil; fluxes of Ca, Mg, and K were about doubled, while fluxes of Na. Cl, SO_4, and N were increased by 4–30 times compared to the pasture. The variations in elemental fluxes were primarily due to differences in solute concentrations rather than differences in the water flux which was 1.5-fold less in the pine forest (512 versus 347 mm of leaching in the pasture and pine, respectively).

Lysimeters versus centrifugation soil solutions

Soil solutions were extracted by centrifugation to determine whether solute concentrations contained in micropores differed appreciably from macropore waters collected by tension lysimeters during the same time period. Soil solutions collected by centrifugation had significantly ($p < 0.05$) higher concentrations of K and NH_4 in both pine and pasture soils, higher concentrations of Cl and SO_4 in pasture soil, and larger concentrations of HCO_3 in pasture soil (Table 3). The large variability associated with soil solution chemistry from both centrifugation and lysimetry collection methods coupled with the low number of replicate samples ($n \leq 4$) contributed to the low number of significant differences between methods.

Soil carbon dioxide concentrations, water content, and temperature

Due to the important role of CO_2 as a proton donor (carbonic acid), pH buffering agent, and mobile anion (HCO_3^-/CO_3^{2-}), concentrations of CO_2 in the soil atmosphere were measured in the year following collection of the soil solutions. No differences were measured between CO_2 concentrations at the 10 and 20 cm depths within either the pasture or pine soils (Figure 6). Carbon dioxide concentrations were more than one order of magnitude greater in the 0–20 cm depth of the pasture soil (35,357 ± 568, μL L^{-1}; mean ± SEM) compared with levels in the pine soil (2,804 ± 308 μL L^{-1}; mean ± SEM). These mean concentrations were 99 and 7.9 times atmospheric levels for the pasture and pine soils, respectively, assuming an atmospheric level of 356 μL L^{-1}.

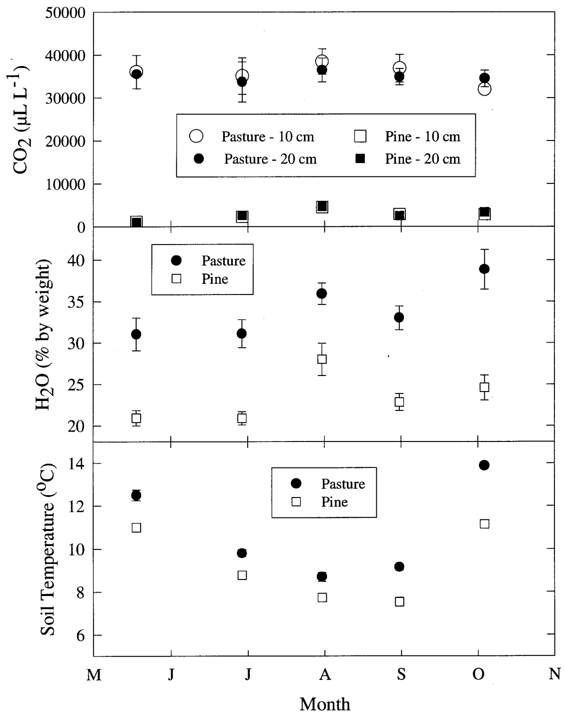

Figure 6. Concentrations (mean ± SEM) of carbon dioxide in the soil atmosphere at the 10 and 20 cm soil depths. Soil water content (mean ± SEM) for the 0–20 cm depth and soil temperature (mean ± SEM) at the 15 cm depth are shown for each site at which the soil atmosphere was collected.

Since soil moisture and temperature have a major role in regulating soil atmosphere CO_2 concentrations through their influence on rates of biological activity and gas diffusion, soil water content and soil temperature were measured at the time of soil gas collection for the 0–20 cm soil depth. Soil moisture content was significantly ($p < 0.05$) greater in the pasture (33.9 ± 1.5%; mean ± SEM) compared with the pine soil (23.5 ± 3.0%; mean ± SEM) throughout the study period (Figure 6). Soil temperatures ranged between 7.5 and 14° C over the study period and averaged 1.6 °C higher in the pasture compared with the pine soil (Figure 6). Thus, the pasture soil was slightly warmer and appreciably wetter throughout the monitoring period.

Discussion

Organic carbon

Several solid-phase soil properties apparently were affected by conversion of pasture to radiata pine. Most

Table 2. Estimated elemental fluxes in the soil solutions passing the 20 cm depth in the pine and pasture sites during the 1995 water year

	Pine	Pasture
	(kg ha^{-1})	
Ca	43	25
Mg	14	8
K	7.8	3.6
Na	145	35
Cl	231	8
N	1.9	0.3
S	25.2	1.7

notable were a redistribution of soil organic carbon, a decrease in total N, soil acidification, and an increase in pools of exchangeable Mg, K, and Na. Organic carbon concentrations and pools decreased throughout the upper 20cm of the mineral soil. However, when the litter layer was included in the organic carbon pool, no difference was exhibited ($p > 0.10$). This indicates that decreases of soil organic matter from the mineral soil had been balanced by gains in the litter layer. During the early stages of stand development in forest ecosystems, there is very little production of detrital materials (e.g. litterfall, root turnover) due to the small biomass and low rates of litterfall return. Residual organic carbon in the mineral soil will be appreciably decreased during this early period of pine development as decomposition continues from root turnover of the dense fibrous rooting system associated with the grass/clover vegetation (Davis, 1995). As canopy closure occurs, higher rates of litterfall production create a surface litter layer that becomes partially incorporated into the upper mineral soil layer by the activity of macrofauna. Thus, the reduction of soil organic matter in the 0–10 cm depth is less than for the 10–20 cm depth increment.

Nitrogen pool

The N pool was 1700 kg ha^{-1} greater (5900 versus 4200 kg ha^{-1} in the upper 20 cm of the mineral soil in the pasture compared with the pine with the litter layer included. Large reductions in the pool of total N have been observed in similar studies of grassland/pasture to pine conversion (Birk, 1992; Davis, 1994, 1995; Davis and Lang, 1991; Giddens et al., 1997).

The large reduction in total N may be attributed to several factors including removal of N fixing clovers, accumulation of N in vegetation biomass, and leaching of N due to enhanced mineralization/nitrification. Differences in the N budget between the two sites include: 70 kg N ha^{-1} of N fertilizer to the pasture during the 1992–96 period, N fixation by clovers in pasture of 50–180 kg N ha^{-1} yr^{-1} (Turvey and Smethurst, 1981), and accumulation of N in biomass within the range of 200 to 400 kg ha^{-1} for 20-year-old radiata pine (Beets and Pallock, 1987; Madgwick, 1985; Stewart et al., 1981). It is also probable that enhanced N mineralization during the early stages of stand development exceeded the plant uptake requirement, resulting in appreciable leaching of NO_3-N. Even at the current stage of stand development, there is the apparent decrease of 2 kg ha^{-1} yr^{-1} from the upper mineral soil (Table 2). The large decrease of N from the upper 20 cm of the mineral soil (1600 kg N ha^{-1}) greatly exceeds the 200–400 kg N ha^{-1} estimated to be contained in the biomass. Thus, leaching of NO_3-N in streamwater may have been appreciable during the early period of stand development (Smethurst and Sadanandan Nambier, 1995).

Soil acidification

Soil acidification by pine was apparent in the upper 20 cm of the mineral soil as demonstrated by both solid-phase and soil solution data. Acidification occurred despite the fact that the soil, a silt loam, had high levels of C, moderate levels of bases and base rich parent material (Alexander and Cresser, 1995). Pine trees may enhance soil acidification through production of organic acids, biomass storage of cations in excess of anions, and canopy capture of acidic pollution (De Vries et al., 1995; Van Breeman et al., 1983). This site, however, receives little atmospheric pollution. The contribution of organic acids to acidification is suggested by the trend for a higher anion charge deficit (primarily due to dissociated organic acids) in the pine soil solutions; however, the difference from the pasture solutions was not significant at p=0.05. Wet and dry deposition is greatly increased by the pine canopy which increases deposition of sea salts (Beier et al., 1993), and Na may exchange with nutrient cations (Figure 1) (Lamersdorf and Meyer, 1993). Radiata pine trees accumulate large concentrations of nutrient cations which will contribute to acidification of the rooting zone (Madgwick, 1985).

A very important acidification process that we were not able to document in this study is the effect of enhanced organic matter decomposition, dissolution of stored S, N mineralization and nitrification associated with the decrease in organic matter from the upper

Table 3. Comparison of solute chemistry (means ± SEM) between samples collected by centrifugation (Cent.) and tension lysimetry (Lysim.). A statistical comparison was made between centrifuged soil solutions at the 10–20 cm depth and the tension lysimeters collected from approximately the 22 cm depth. The pH of the lysimeter solutions was corrected for the effect of CO_2 degassing

Method	Depth (cm)	pH	Ca	Mg	K	Na	NH_4	HCO_3	Cl	NO_3	SO_4
			(m*M*)								
Pasture											
Cent.	0–10	5.53 (0.03)	0.34 (0.04)	0.14 (0.01)	0.10 (0.01)	0.45 (0.03)	0.05 (0.01)	<0.01	1.06 (0.04)	0.01 (<0.01)	0.07 (0.01)
Cent.	10–20	5.91 (0.13)	0.15 (0.01)	0.06 (0.01)	0.05 (0.01)*	0.41 (0.06)	0.04 (0.01)*	<0.01	0.47 (0.08)*	0.02 (<0.01)	0.04 (0.01)*
Lysim.	22	6.28 (0.12)	0.13 (0.03)	0.06 (0.01)	0.02 (<0.01)*	0.31 (0.02)	<0.01*	0.32 (0.07)*	0.06 (0.04)*	0.01 (<0.01)	<0.01*
Pine											
Cent.	0–10	5.35 (0.19)	0.32 (0.05)	0.19 (0.05)	0.24 (0.02)	1.26 (0.26)	0.12 (0.05)	<0.01	1.53 (0.44)	0.32 (0.10)	0.10 (0.02)
Cent.	10–20	5.30 (0.32)	1.08 (0.69)	0.71 (0.46)	0.29 (0.06)*	3.61 (1.29)	0.09 (<0.01)*	<0.01	6.74 (5.22)	0.09 (0.03)	0.12 (0.05)
Lysim.	22	6.02 (0.07)	0.30 (0.05)	0.18 (0.04)	0.05 (0.01)*	1.76 (0.48)	<0.01*	0.01 (<0.01)	1.93 (0.71)	0.05 (0.02)	0.26 (0.05)

*Differences between means (centrifuge versus lysimeter) are statistically different at $p = 0.05$ as determined by a t-test.

20 cm of the mineral soil (Lamersdorf and Meyer, 1993). Production of protons by nitrification and subsequent leaching of base cations with the NO_3-N may have contributed to soil acidification during the early stage of stand development when N uptake requirements were low. This response is often observed in forest soils following clearcutting when organic matter decomposition is enhanced (Dahlgren and Driscoll, 1994).

Pools of cations

The pools of exchangeable Mg, K, and Na were increased following conversion of pasture to pine. We believe that two processes are primarily responsible for this observation. First, nutrient cycling by the pine results in absorption of nutrients from lower soil horizons and their subsequent return to the soil surface via litterfall and canopy throughfall. Since pine roots are distributed much deeper in the soil profile than the shallow rooted grasses and clovers in the pasture, nutrients may be retained more strongly in the pine ecosystem. Second, increased dry deposition and fog/cloud deposition by the pine canopy results in much greater inputs of elements associated with sea salts, and this is further enhanced at the forest edge or in small forests (Beier et al., 1993). In particular, Na and Mg along with the Cl and S anions are enriched. As stated earlier, there was no difference in the pool of exchangeable Ca which arises in part from the historic application of superphosphate fertilizer to the pasture.

Soil solution chemistry and fluxes

The concentrations of ions in soil solution and stream runoff were considerably higher in the pine compared with the pasture (Figures 1–3 and Table 2). The primary control on ion leaching is the availability of mobile anions (Johnson and Cole, 1980). The dominant anions in the pine soil solutions were Cl and SO_4, which were very low in the pasture solutions. This large difference between the two sites may be explained by enhanced deposition of these anions by canopy capture of dry deposition (e.g. aerosols and particulate forms) and fog/cloud deposition having a high sea salt component (Beier et al., 1993). The mean Na/Cl ratio for the pine soil solutions (1.07) was nearly identical to that of sea water (1.17), providing strong support for the sea salt origin of these solutes (Stumm and Morgan, 1970). Sea salts appear to accumulate in the upper soil horizons of the pine soil during the summer months when precipitation amounts are sufficient to wash off the canopy but not enough to initiate leaching. In contrast, the lack of appreciable above-ground biomass in the pasture results in little capture of sea salts associated with dry deposition and fog/cloud deposition. A small increase in the concentrations of Na, K, and Cl in the first soil solution collection of the season from the pasture may reflect a small flush of sea salts from the upper mineral soil. With respect to surface water quality, solute concentration and fluxes are expected to be considerably greater in watersheds draining pine forests compared with pasture vegetation.

With the exception of HCO_3 and NO_3, all major solutes in the pine soil solutions show a progressive decrease with time (Figures 2 and 3). We believe that this pattern results from the progressive leaching of salts from micropores. Salts originating from dry deposition and fog/cloud deposition would be expected to become concentrated in the micropores during the summer period of desiccation. These salts begin leaching from the micropores with the onset of macropore flow during the winter. It appears that several pore volumes are required to effectively flush micropore solute concentrations to levels comparable with those found in canopy throughfall inputs to the soil. The generally higher salt concentrations measured in the micropores (centrifuged soil solutions) compared with the macropores (lysimeter soil solutions) during the month of August support this mechanism (Table 3). We would expect the differences in solute concentrations between centrifuged and lysimeter soil solutions to be even greater at the onset of the rainy season. It is not clear whether preferential flowpaths or non-equilibrium between macropores and micropores contribute to differences in solute concentrations between these two types of soil solutions.

The dominant inorganic anion in the pasture soil solutions and stream runoff was HCO_3. Mean HCO_3 concentrations were 4-fold greater in the pasture compared with the pine. This difference is primarily due to differences in CO_2 concentrations of the soil atmosphere. Carbon dioxide concentrations were 12.5-fold greater in tree pasture soil compared with the pine in the year following the collection of soil solution (Figure 6). In addition, the slightly greater soil pH (0.3–0.4 units) in the pasture leads to a greater dissociation of carbonic acid in the pasture compared with the pine (Table 1 and Figure 4). The greater CO_2 concentrations in the soil atmosphere of the pasture are believed to result from greater microbial and root respiration rates and lower gas diffusion rates in the pasture. The grass/clover vegetation has a dense mass of roots extending to a depth of approximately 20 cm, but especially concentrated in the upper 10 cm. In contrast, there were no understory species in the pine site and relatively few fine roots compared with the pasture. Soil organic matter concentrations were greater and the C/N molar ratio slightly lower in the pasture site, both of which would tend to favour enhanced microbial respiration (decomposition). Similarly, the warmer soil temperature (1.6 °C greater) in the pasture would enhance soil respiration rates. While the pasture has a greater porosity based on bulk density measurements (60 versus 55%), gas diffusion rates are believed to be considerably lower in the pasture soils due to the higher moisture content (10% greater) throughout the winter leaching season. The net effect of these contributing factors is much greater CO_2 and HCO_3 concentrations in the pasture, leading to bicarbonate being the dominant inorganic anion regulating leaching fluxes.

In contrast to the high HCO_3 concentrations measured in soil solutions collected by lysimetry, HCO_3 concentrations in soil solutions obtained by centrifugation were less than detection limits (<0.01 m*M*). A previous study showed that these low bicarbonate concentrations resulted from degassing of the solid-phase soil samples during the two hour storage/transport period before solution extraction by centrifugation (Dahlgren et al., 1997). In addition to the lower bicarbonate concentrations, cation concentrations in centrifuged soil solutions were underestimated by an amount equivalent to the loss of the bicarbonate anion.

Concentrations and fluxes of N, P, and S in soil solutions were very low in the pasture, in spite of the long-term application of these nutrients with annual fertilization, and these data were consistent with those of Sakadevan et al. (1993) for grazed hill country pastures. Fertilizer additions of these three nutrients commonly provide enhanced forage yields indicating that these nutrients are deficient in hill country pasture. In contrast, soil solution concentrations and fluxes of S, and to a much smaller degree N, were enhanced in the pine soil that received no current or historic additions of fertilizer. The elevated N and S fluxes may result, in part, from increased rates of mineralisation and organic matter decomposition as suggested by the solid-phase data from this study and several studies from the literature (Birk, 1992; Davis and Lang, 1991; Davis, 1994, 1995). However, the largest source of S is believed to originate from sea salts captured by the pine canopy.

Conclusions

The solid-phase data indicate that there is a decrease in organic C from the upper mineral soil when land in pasture is converted to radiata pine; however, this decrease is largely compensated for by accumulation of organic C in the litter layer under the pine. There is also a decrease of soil N and an increase in soil acidification in the upper mineral soil under pine. These results are entirely consistent with previous short-term studies (Birk, 1992; Davis, 1994, 1995; Davis and Lang, 1991) and with long-term (17–30 years) comparisons from 10 paired pasture/pine sites in New Zealand (Giddens et al., 1997).

Nutrient concentrations and fluxes in soil solutions were considerably greater in the pine compared with the pasture soils due to differences in the availability of mobile anions. The dominant anion under pasture was HCO_3 resulting from the high CO_2 concentrations in the soil atmosphere under pasture vegetation. In contrast, Cl and SO_4 were the dominant anions in the pine soil solutions, resulting from capture of sea salts by the pine canopy. The removal of the tree canopy substantially reduces sea salt deposition and probably contributes to the S deficiency experienced in some hill country pastures. In spite of the higher leaching losses of nutrients from the upper soil horizons of the pine soil, nutrient pools are generally sustained due to nutrient cycling by the pine and enhanced capture of sea salts by the pine canopy.

Acknowledgements

We are grateful to A D Mackay, J Napier, and D Barker of AgResearch, New Zealand for access to the sites, for their valuable assistance, and for the rainfall data; and to K Giddens of Landcare Research for the solid-phase soil analyses. This work was supported under Foundation for Research, Science and Technology contract C09520.

References

Alexander C E and Cresser M S 1995 An assessment of the possible impact of native woodland cover on the chemistry of Scottish freshwaters. For. Ecol. Manage. 73, 1–27.

Beets P N and Pollock D S 1987 Uptake and accumulation of nitrogen in *Pinus radiata* stands as related to age and thinning. N.Z. J. For. Sci. 17, 353–371.

Beier C, Hansen K and Gundersen P 1993 Spatial variability of throughfall fluxes in a spruce forest. Environ. Pollut. 81, 257–267.

Birk E M 1992 Nitrogen availability in radiata pine plantations on former pasture sites in southern New South Wales. Plant Soil 143, 115–125.

Blakemore L C, Searle P L and Daly B K 1987 Methods for chemical analysis of soils. N.Z. Soil Bureau Scientific Report 80. Dept. Scientific and Industrial Research, Lower Hutt.

Cole D W and Gessel S P 1965 Movement of elements through forest soil as influenced by tree removal and fertilizer addition. *In* Forest-Soil Relationships in North America. Proc. and North American Forest Soil Conference, Corvallis, Oregon, August 1963. Ed. C T Youngberg. pp 95–104. Oregon State Univ. Press, Corvallis.

Condron J 996 Trees and soil fertility: A New Zealand perspective. *In* Proceedings ASSSI and NZSS National Soils Conference-Oral Papers, Melbourne, July 1996. pp 47–48. ASSI, NZSS, Melbourne.

Dahlgren R A and Ugolini F C 1989 Effects of tephra addition on soil processes in spodosols in the Cascade Range, Washington, USA. Geoderma 45, 331–355.

Dahlgren R A and Driscoll C T 1994 The effects of whole-tree clear-cutting on soil processes at the Hubbard Brook Experimental Forest, New Hampshire, USA. Plant Soil 158, 239–262.

Dahlgren R A, Percival H J and Parfitt R L 1997 Carbon dioxide degassing effects on soil solutions collected by centrifugation. Soil Sci. (*In press*).

Dahlgren R A, Ugolini F C, Shoji S, Ito T and Sletten R S 1991 Soil-forming processes in Alic Melanudands under Japanese pampas grass and oak. Soil Sci. Soc. Am. J. 55, 1049–1056.

Davis M R 1994 Impact of afforestation on a tussock grassland soil in Otago. N.Z. J. Agric. Res. 37, 465–469.

Davis M R 1995 Influence of radiata pine seedlings on chemical properties of some New Zealand montane grassland soils. Plant Soil 176, 255–262.

Davis M R and Lang M H 1991 Increased nutrient availability in topsoils under conifers in the South Island high country. N.Z. J. For. Sci. 21,165–179.

De Vries W, van Grinsven J J M, van Breemen N, Leeters E E J M and Jansen P C 1995 Impacts of acid deposition on concentrations and fluxes of solutes in acid sandy forest soils in the Netherlands. Geoderma 67, 17–43.

Giddens K M, Parfitt R L and Percival H J 1997 Comparison of some soil properties under *Pinus radiata* and pasture. N.Z. J. Agric. Res. (*In press*).

Heng L K, White R E, Bolan N S and Scotter D R 199l. Leaching losses of major nutrients from a mole-drained soil under prasture. N.Z. J. Agric. Res. 34, 325–334.

Hewitt A E 1992 New Zealand Soil Classification. DSIR Land Resources Scientific Report No. 19. N.Z. Dept. Scientific and Industrial Research, Lower Hutt.

Johnson D W and Cole D W 1980 Anion mobility in soils: Relevance to nutrient transport from terrestrial ecosystems. Environ. Int. 3, 79–80.

Lamersdorf N P and Meyer M 1993 Nutrient cycling and acidification of a northwest German forest site with high atmospheric nitrogen deposition. For. Ecol. Manage. 62, 323–354.

Maclaren J P 1996 Environmental Effects of Planted Forests in New Zealand. FRI Bulletin No.198. New Zealand Forest Research Institute, Rotorua. 18 p.

Madgwick H A I 1985 Dry matter and nutrient relationships in stands of *Pinus radiata*. N.Z. J. For. Sci. 15, 324–336.

Mead D J 1995 The role of agroforestry in industrialised nations: the southern hemisphere perspective with special emphasis on Australia and New Zealand. Agrofor. Syst. 31, 143–156.

New Zealand Meteorological Service 1983 Summaries of climatological observations to 1980. N.Z. Met. Service Misc. Publ. 177, Wellington.

Rustad L E, Fernandez I J, Fuller R D, David M B, Nodvin S C and Halteman W A 1993 Soil solution response to acidic deposition in a northern hardwood forest. Agric. Ecosyst. Environ. 47, 117–134.

Sakadevan K, Hedley M J and Mackay A D 1993 Mineralisation and fate of soil sulphur and nitrogen in hill pastures. N.Z. J. Agric. Res. 36, 271–281.

Smethurst P J and Sadanandan Nambier E K 1995 Changes in soil carbon and nitrogen during the establishment of a second crop of *Pinus radiata*. For. Ecol. Manage. 73, 145–155.

Soil Survey Staff 1994 Keys to Soil Taxonomy (6th edition). SMSS Tech. Monogr. No.19. Pocahontas Press, Blacksburg, VA.

Stevenson C D 1978 Simple apparatus far monitoring land disposal systems by sampling percolating soil waters. Environ. Sci. Technol. 12, 329–331.

Stevenson C D and Beaumont H M 1984 Monitoring techniques for land application of effluents. *In* Land Treatment of Wastes, Ed. Wilcock. pp 387–408. Water and Soil Miscellaneous Publication No. 70. National Water and Soil Conservation Authority, Wellington.

Stewart H T L, Flinn D W and James J M 1981 Biomass and nutrient distribution in radiata pine. *In* Proceedings of Australian Forest Nutrition Workshop ‘Productivity in Perpetuity’, Canberra, Australia. August, 1981. pp 173–185. CSIRO, Melbourne.

Stumm W and Morgan J J 1970 Aquatic Chemistry - An Introduction Emphasizing Chemical Equilibria in Natural Waters. Wiley Interscience, New York. 583 p.

Suarez D L 1987 Prediction of pH errors in soil-water extractors due to degassing. Soil Sci. Soc. Am. J. 51, 64–67.

Turvey N D and Smethurst P J 1981 Economic and biological criteria for establishing nitrogen fixers in pine plantationis. *In* Managing Nitrogen Economics of Natural and Man-made Forest Ecosystems. Eds. R A Rummey and F J Hingston. pp 124–145. CSIRO, Melbourne.

Van Breemen N, Mulder J and Driscoll C T 1983 Acidification and alkalinization of soils. Plant Soil 75, 283–308.

Vitousek P M, Matson P A and Van Cleve K 1989 Nitrogen availability and nitrification during succession: Primary, secondary and old-field seres. Plant Soil 115, 229–239.

Section editor: R F Huettl